

Scalability & Future Plan

Date	04/07/2026
Team ID	xxxx
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the Personalized Networking Assistant can be scaled beyond its current single-user, local-storage design.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Local JSON storage doesn't support concurrent multi-user writes	Data conflicts if used by more than one person at once	High
2	Gemini free-tier API has a limited request quota	Starter generation may fall back to templates during heavy testing/usage	Medium
3	No authentication/authorization layer	Not suitable for multi-user/public deployment as-is	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user	Deploy behind FastAPI + Uvicorn workers (or Gunicorn) to support multiple concurrent users
2	Data Storage	Local JSON files (history.json, feedback.json)	Migrate to PostgreSQL/MongoDB for concurrent, multi-user data access
3	Performance	Sequential Gemini/Wikipedia calls per request	Add caching for repeated event themes/fact-checks and async I/O for API calls
4	Security	No authentication; API key stored in local .env	Add user authentication (JWT/OAuth2) and move secrets to a managed secret store before public hosting

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add basic user accounts and login	Q3 2026	Enables personalized history per user
Phase 3	Migrate storage to a hosted database (PostgreSQL/MongoDB)	Q4 2026	Supports multiple concurrent users reliably

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 4	Deploy to a cloud host (e.g., Streamlit Cloud / Render) with CI/CD	Q1 2027	Makes the assistant publicly accessible and always-on